

TPU specs

Here are some specific numbers for our chips:

Model	Pod size	Host size	HBM capacity/chip	HBM BW/chip (bytes/s)	FLOPs/s/chip (bf16)	FLOPs/s/chip (int8)
TPU v3	32x32	4x2	32GB	9.0e11	1.4e14	1.4e14
TPU v4p	16x16x16	2x2x1	32GB	1.2e12	2.75e14	2.75e14
TPU v5p	16x20x28	2x2x1	96GB	2.8e12	4.59e14	9.18e14
TPU v5e	16x16	4x2	16GB	8.1e11	1.97e14	3.94e14
TPU v6e	16x16	4x2	32GB	1.6e12	9.20e14	1.84e15

Host size refers to the topology of TPUs connected to a single host (e.g. TPU v5e has a single CPU host connected to 8 TPUs in a 4x2 topology). Here are interconnect figures:

Model	ICI BW/link (one-way, bytes/s)	ICI BW/link (bidi, bytes/s)
TPU v3	1e11	2e11
TPU v4p	4.5e10	9e10
TPU v5p	9e10	1.8e11
TPU v5e	4.5e10	9e10
TPU v6e	9e10	1.8e11

PCIe bandwidth is typically around 1.6×10^{10} bytes / second per TPU (3.2×10^{10} for TPU v6e), while DCN bandwidth is typically around 6.25×10^9 bytes / second per TPU (12.5×10^9 for TPU v6e and 3.125×10^9 for TPU v5e).

Question 1 [bounding LLM latency]: Say you want to sample from a 200B parameter model in bf16 that's split across 32 TPU v4p. How long would it take to load all the parameters from HBM into the systolic array? *Hint: use the numbers above.*

my answer:-

size to load= $200 \times 10^9 \times 2$ Bytes

total time=size/(32*HBM speed)=0.01 sec = 10 ms

Question 2 [TPU details]: Consider a full TPU v5e pod. How many total CPU hosts are there? How many TPU TensorCores? What is the total FLOPs/s for the whole pod? What is the total HBM? Do the same exercise for TPU v5p pod.

My Answer:-

pod size=16x16 chips

Hosts=4x2=9

TensorCores=numchips=256

computation= TC*1.97e14=5.04e16

HBM=chips*16=4096GB

TPU v5p:-

pod size=16x20x28 chips

Hosts=2x2x1=4

TensorCores=numchips=8960

computation=TC*4.56e14=4.11e18

HBM=chips*96=~860TB

Question 3 [PCIe operational intensity]: Imagine we're forced to store a big weight matrix A of type bfloat16 $[D, F]$, and a batch of activations x of type bfloat16 $[B, D]$ in host DRAM and want to do a matrix multiplication on them. This is running on a single host, and we're using a single TPU v6e chip attached to it. You can assume $B \ll D$, and $F=4D$ (we'll see in future chapters why these are reasonable assumptions). What is the smallest batch size B we need to remain FLOPs bound over PCIe? Assume PCIe bandwidth of 1.5e10 bytes / second.

16x20x

My Answer:-

$AI = 2 \cdot BDC / (2 \cdot (BD + DF + BF)) > 9.20e14 / 1.5e10$

$B > 6133$

Question 4 [general matmul latency]: Let's say we want to multiply a weight matrix int8[16384, 4096] by an activation matrix of size int8[B, 4096] where B is some unknown batch size. Let's say we're on 1 TPU v5e to start.

1. How long will this multiplication take as a function of B? *Hint: it may help to calculate how long it will take to load the arrays from HBM and how long the multiplication will actually take. Which is bottlenecking you?*
2. What if we wanted to run this operation out of VMEM? How long would it take as a function of B?

My Answer:-

$$T_{\text{comp}} = 2 \cdot B \cdot D / 3.94 \times 10^{14}$$

$$T_{\text{comm}} = B \cdot D + B \cdot F / 8.1 \times 10^{11}$$

$$T = \max(T_{\text{comp}}, T_{\text{comm}})$$

it is computation bound if $T_{\text{comp}} > T_{\text{comm}}$ $T = B / 2.93 \times 10^6$ for $B > 250$ around
and else $T = (6.7 \times 10^7 + 2 \times 10^4 B) / 8.1 \times 10^{11}$

2) let say vmem BW is 20 times of HBM then B will be reduced by 20 time $B > 12$

Question 5 [ICI bandwidth]: Let's say we have a TPU v5e 4×4 slice. Let's say we want to send an array of type `bfloat16[8, 128, 8192]` from `TPU{0, 0}` to `TPU{3, 3}`. Let's say the per-hop latency for TPU v5e is $1 \mu\text{s}$

1. How soon will the first byte arrive at its destination?
2. How long will the total transfer take?

My Answer:-

1) it will pass through 6 ICI links so 1st byte will reach at around 6 micro s

2) very very very interesting we can send array through 2 paths and final chip can receive it through 2 parts so essentially the speed doubled
num of bytes = $2 \times 8 \times 128 \times 8192 = 1.7 \times 10^7$
and the number of paths is 2 and its happening simultaneously so no 6 multiplier so speed is time is $1 \times 10^6 (\text{micro sec}) \times 1.7 \times 10^7 / (2 \times 4.5 \times 10^{10}) = 189 \text{ micro sec}$

Question 6 [pulling it all together, hard]: Imagine you have a big matrix A: `int8[128 * 1024, 128 * 1024]` sharded evenly across a TPU v5e 4×4 slice but offloaded to host DRAM on each chip. Let's say you want to copy the entire array to `TPU{0, 0}` and multiply it by a vector `bfloat16[8, 128 * 1024]`. How long will this take? *Hint: use the numbers above.*

My Answer:-

$$A = 128 \times 1024 \text{ bytes}, A \times 2 = 16 \text{ GB of mem}$$

Copy sharded to chip HBM from host Dram= $16\text{GB}/(16 \text{ chips} * \text{DCN})=66\text{ms}$
copy 2nd array from HBM to Mxu= $2*8*128*1024/8.1\text{e}11 = 2.59 \text{ micro sec}$
copy large array from HBMs to same TPU= $15\text{GB through bi iCI path}=1.5/9=167 \text{ ms}$
copy large array from TPU HBM to tensor= $16\text{e}9/8.1\text{e}11=19\text{ms}$
 $T_{\text{comp}}=16*16*\text{e}9/T_{\text{for bf16}}=1.3 \text{ ms}$